

Which Error Can the District Afford?

AP Statistics · Topic 6.7 (CED 3.8) · B: 2026-12-08 · E: 2027-01-15

[STAR] TODAY'S PROBLEM

A district tests $H_0 : p = 0.05$ versus $H_a : p > 0.05$, where p is the late-alert proportion. Both plans use $n = 200$ and have the table's operating probabilities. Safety prioritizes avoiding retention when $p = 0.10$; Budget prioritizes avoiding replacement when $p = 0.05$. No common cost scale exists.

Dev: "Use A. Smaller α lowers both error risks."

Lin: "Use B. Its higher power has no cost when safety matters."

Test plans ($n = 200$)

Plan	α	Power at $p = 0.10$
A	0.010	0.748
B	0.050	0.877

FIRST TAKE — before the video (5 min)

Which plan would you try first? Name the office you prioritized and one competing cost.

[BOOK] ERROR PROBABILITIES TRADE OFF (keep on the page)

A Type I error rejects a true H_0 , and its probability is α . A Type II error fails to reject a false H_0 ; $\beta = 1 - \text{power}$. With sample size fixed, increasing α generally raises power and lowers β , but it also raises the false-positive risk. Increasing sample size can raise power while keeping the chosen α fixed. Choose a plan by naming the real consequence of each error.

Finish after the video:

DOK 1 (a) Describe a Type I error and a Type II error in this context. Identify what $P(\text{reject } H_0 \mid p = 0.10)$ represents.

DOK 2 (b) For each plan, calculate β . Then give the expected number of Type I errors per 1,000 repeated tests when $p = 0.05$ and Type II errors per 1,000 repeated tests when $p = 0.10$.

DOK 3 (c) Adjudicate both claims. Explain why no plan is universally best; recommend one for a stated office using both error probabilities and their per-1,000 consequences. Name information needed for a district-wide choice and a change that improves power at fixed α .

Frame: I recommend Plan _____ because its chance of missing a 10 percent late rate is _____, compared with _____.

Frame: That benefit costs about _____ additional _____ per 1,000 tests; increasing _____ could improve power while holding α fixed.

EXIT — turn this in

One thing the video changed about my first take: _____